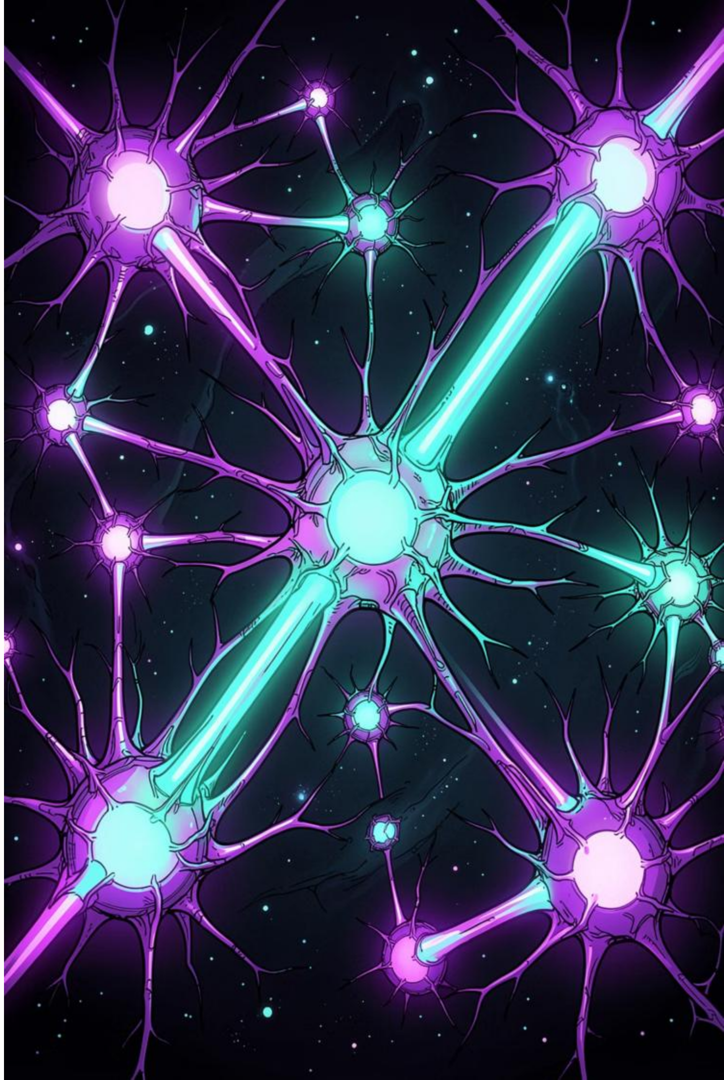
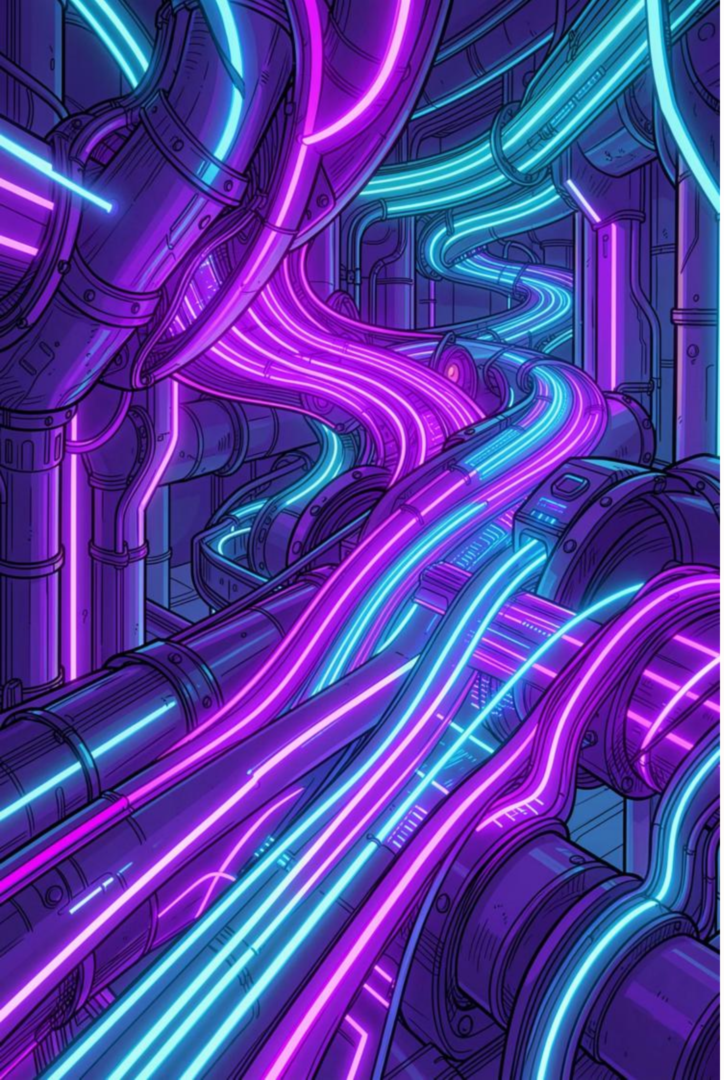
HACKATHON SUBMISSION

India Runs Data & AI Challenge

AI-Powered Candidate Ranking System

Intelligent, scalable, and transparent — ranking 100,000 profiles to surface the best Senior AI Engineer candidates.





PROBLEM STATEMENT

The Ranking Challenge

Given a large-scale candidate dataset, the goal is to intelligently rank and shortlist the most qualified profiles for a **Senior AI Engineer** role — with precision, speed, and a clean, reproducible output.

100,000 Profiles

Process a massive candidate dataset at scale

JD Matching

Align candidates to a Senior AI Engineer job description

Top 100 Output

Generate a ranked shortlist of the best-fit candidates

Valid submission.csv

Produce a clean, submission-ready CSV file

DATA

Dataset Overview

Eight rich data dimensions power the ranking model — each contributing unique signals to evaluate candidate quality.

Candidate Profiles

Core identity, location, and professional summary

Skills

Technical competencies and domain expertise tags

Career History

Work experience, roles, and tenure data

Education

Degrees, institutions, and academic background

Certifications

Professional and cloud certifications earned

Redrob Signals

Platform-specific recruiter engagement metrics

GitHub Activity

Commits, repos, and open-source contributions

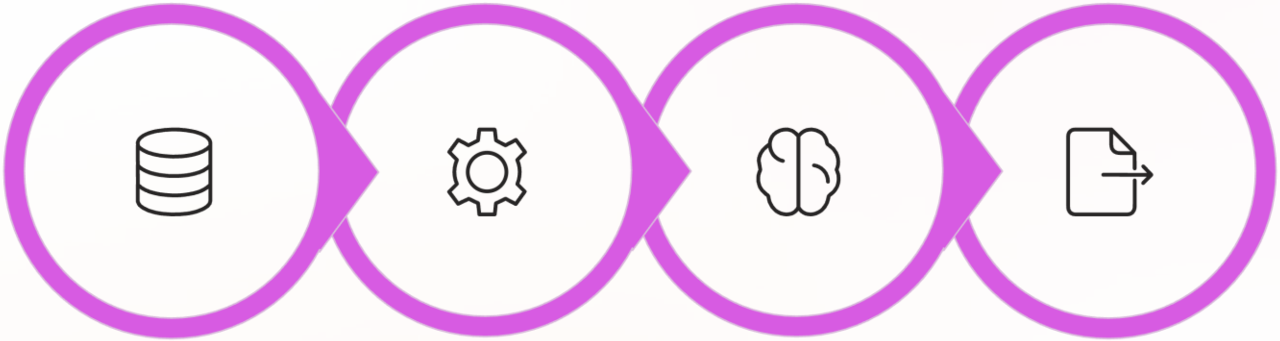
Recruiter Metrics

Response rates, profile views, and outreach data

ARCHITECTURE

Solution Architecture

A modular, end-to-end pipeline transforms raw candidate data into a ranked, submission-ready output — with every stage optimised for speed and accuracy.



Data Ingestion

Feature Engineering

AI Scoring

Output

Each stage is independently testable, enabling rapid iteration and transparent debugging during the hackathon sprint.

TECH STACK

Technologies Used

Python

Core language for data processing, scoring logic, and pipeline orchestration

Pandas

High-performance data manipulation, filtering, and aggregation at scale

JSON

Structured data parsing for nested skill and profile fields

Google Colab

Cloud-based execution environment with GPU access for rapid prototyping

GitHub

Version control, collaboration, and public repository for reproducibility

ALGORITHM

Ranking Algorithm

Candidates are scored across **six weighted dimensions**, each reflecting a critical signal of Senior AI Engineer readiness. The final composite score determines the ranking order.

Experience (5–9 Years)

Optimal seniority band for the role

AI Skills

Python, NLP, LLMs, RAG, Vector DB

Recruiter Signals

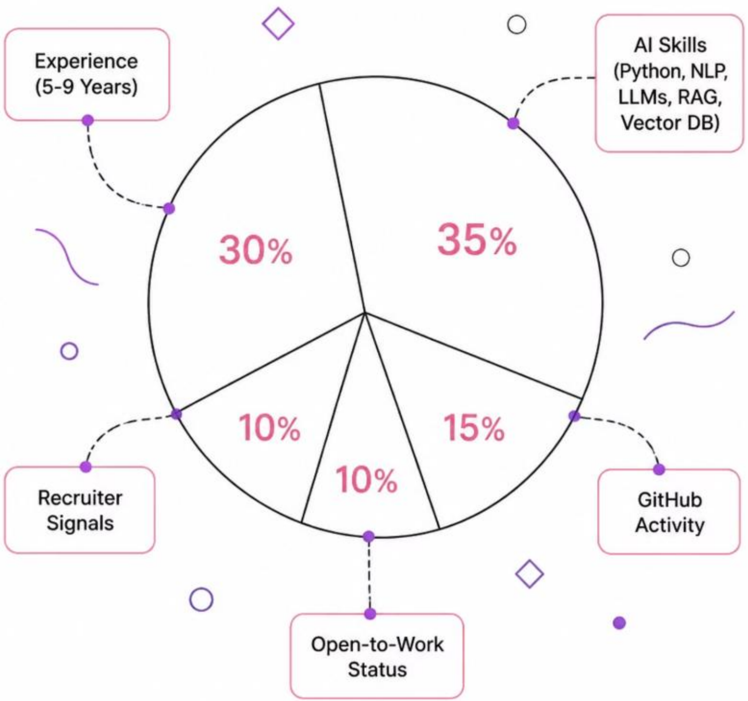
Engagement and response metrics

GitHub Activity

Code quality and open-source presence

Open-to-Work

Immediate availability signal



RESULTS

Processing Results

100K

Profiles Processed

Full dataset ingested, cleaned, and scored

100

Candidates Ranked

Top 100 shortlist generated with composite scores

1

submission.csv

Valid, submission-ready output file produced

~0s

Processing Time

Optimised pipeline for fast, scalable execution

WHAT'S NEXT

Advantages & Future Improvements

✅ Current Advantages

- **Fast & Scalable** — Handles 100K profiles efficiently
- **Transparent Scoring** — Interpretable, rule-based weights
- **AI-Focused** — Tailored specifically for AI Engineer roles

🚀 Future Improvements

- **Semantic Embeddings** — Vectorise skills and JD text
- **Resume–JD Similarity** — Cosine similarity scoring
- **LLM-Based Ranking** — Generative AI for nuanced evaluation
- **ML Ranking Models** — Learn weights from historical hires

REPOSITORY

GitHub Repository Structure

The project follows a clean, modular structure — making it easy to audit, extend, and reproduce. Every file has a clear purpose.

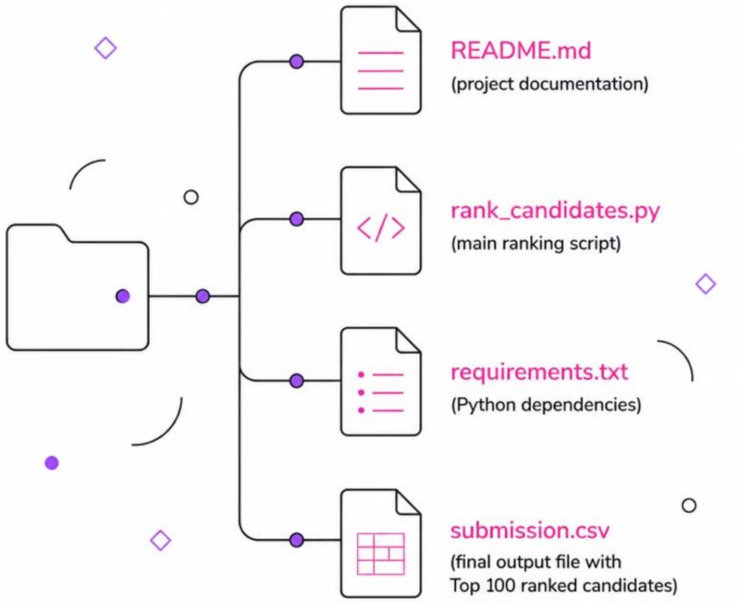
📁 Project Root
Organised file hierarchy for clarity

📄 README.md
Setup instructions, usage guide, and methodology

📊 submission.csv
Final ranked output — Top 100 candidates

rank_candidates.py
Core ranking logic and scoring pipeline

📦 requirements.txt
Reproducible environment with pinned dependencies



Thank You

We built an intelligent, transparent, and scalable candidate ranking system — purpose-built for the India Runs Data & AI Challenge.

🙋 Questions?

Happy to walk through the architecture, scoring logic, or results in detail.

📬 Contact

[Your Email] · [Your LinkedIn] · [GitHub Repository Link]

India Runs Data & AI Challenge — AI-Powered Candidate Ranking System · Hackathon Submission